Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/23/2014
PI Name:	Delp, Michael Ph.D.		
Project Title:	Effects of Microgravity on Cerebral Arterial, Venous, and Lymphatic Function: Implications for Elevated Intracranial Pressure		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	None		
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City:	Tallahassee	**State:**	FL
Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	06/20/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 6/20/2014 (from 8/31/2016) per PI move to Florida State University. Project will continue under a new grant. Info per A. Chu/ARC and NSSC (Ed., 9/23/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Behnke, Brad (University of Florida) Zawieja, David (Texas A&M University)		
Grant/Contract No.:	NNX13AN33G		
Performance Goal No.:			
Performance Goal Text:			

	Approximately 29% of astronauts on short-term (~2 wk) space shuttle flights and 60% on long-duration (~6 mo) missions to the International Space Station (ISS) are reported to have experienced some impairment in distant or near visual acuity. These visual disturbances have been hypothesized to be related to increases in intracranial pressure. Increases in intracranial pressure associated with spaceflight are thought to occur through the cephalad shift of intravascular and cerebrospinal fluids due to the loss of the head-to-foot gravity vector present on Earth. In regard to the vascular component, increases in cerebral blood flow and capillary fluid filtration could elevate fluid movement into the cranium, while impaired function of cerebral veins and lymphatics could impede intracranial fluid drainage out of the cranium. Therefore, the purpose of this proposal is to examine the effects of long-term spaceflight on the vasoconstrictor, mechanical, and structural properties of cerebral arteries, veins, and lymphatic vessels. Flight: The aim of this project is to investigate whether spaceflight on the ISS alters the blood-brain barrier in rodents, as indicated by ultrastructural examination of the cerebral capillary endothelium. To assess the changes that occur in the structure of the blood-brain barrier, 10-15 female C57BL/6 mice, 10-11 weeks old will be loaded into an transporter (AEM-T; 10 mice/side), and late-loaded into a SpaceX Dragon rocket. After ascent and docking to the ISS, the mice will be transferred via an Animal Access Unit (AAU) and Mouse Transfer Box (MTB) to 2 Habitats (AEM-X). The mice will be grouped into four experimental groups, each housed in one side of an AEM-X. The groups, as determined by another investigation, will consist of mice receiving intraperitoneal (i.p.) injection of saline vehicle (n=5), Tetanus Toxoid (TT) vaccine (n=5), CpG ODN adjuvant (n=5), or TT vaccine + CpG ODN adjuvant (n=5).
Task Description:	After mice have been exposed to microgravity for a minimum of 2 weeks, animals for this study will be inoculated with an injection of phosphate buffered saline (n=5) or CpG ODN adjuvant (n=5); a third group of animals injected with the TT vaccine (n=5) may be utilized. Two weeks later (14 +/- 2 days), the mice will be sacrificed in the Materials Science Glovebox (MSG). Sacrificing will be achieved via i.p. injection of 100/10 mg/kg ketamine/xylazine anesthesia followed by exsanguination by cardiac puncture. Following blood collection and harvest of spleen for another research investigation, the animals will have heads removed. The top of the skull will be cut along the sagittal crest and suture and retracted back to expose the brain. The brains will be will be dissected from the mice and put in a solution of 10% buffered formalin at ambient temperature. 24-48 hours later, the 10% buffered formalin will be exchanged with the Brain Fixation Kit (BFK) to 70% Ethanol and stored at ambient temperature until they can be returned to Earth and analyzed. Ground: 20-25 (saline vehicle (n=10), Tetanus Toxoid (TT) vaccine (n=5), CpG ODN adjuvant (n=10)) ground control mice will be put in AEMs inside Orbiter Environmental Simulators (OES) on a 2-8 day delay, depending on the when telemetry data from the SpaceX Dragon and ISS environments are available. The differences in vascular structure between ground controls and flight animals with a P-value < 0.05 will be considered statistically significant. EDITOR's NOTE (9/23/2014): PI moved to Florida State University in June 2014; new continuation grant will be awarded.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cerebral arteries are a widely studied segment of the circulation, particularly in various pathological conditions, such as in hypertension and stroke. In contrast, very little is known regarding the cerebral venous circulation, and even less regarding cranial lymphatic vessels in the role they play in cerebral fluid homeostasis. The present studies will expand our understanding of how these vessels function, and more importantly, how these vessels adapt to environmental stress. One of the most significant aspects of these experiments is that adaptations of these distinct segments of the cerebral circulation will be studied in unison rather than as isolated experiments. To our knowledge, this will be the first time these vascular segments will be studied simultaneously to provide a more comprehensive understanding of factors that impact cerebral fluid dynamics. Such an approach may provide new insight into how these vascular segments adapt together under various conditions on Earth.
Task Progress:	EDITOR's NOTE (9/23/2014): PI moved to Florida State University in June 2014; new continuation grant will be awarded. New grant number will be posted when available. UPDATE (8/3/2016): Continuation grant is "Effects of Microgravity on Cerebral Arterial, Venous, and Lymphatic Function: Implications for Elevated Intracranial Pressure--NNX16AC28G," with period of performance of 12/1/2015-11/30/2018. See that task for continued progress.
Bibliography Type:	Description: (Last Updated: 07/09/2025)